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Conceptualising heart disease prediction through a unified framework combining clinical theory and machine learning models

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Abstract

A hybrid framework for a comprehensive evaluation of machine learning and transformer-based models for cardiovascular risk prediction by integrating structured clinical data with unstructured clinical narratives is included in this study. The structured data includes demographic and diagnostic variables like patient information and test results, while unstructured data is derived from electronic health records such as clinical notes. Through preprocessing, feature engineering, and semantic embedding using transformer-based models, the system leverages the complementary strengths of both data types. By analysing and processing this unstructured information, this project aims to improve the predictions for heart disease. The results show that it is feasible to make better predictions and gain valuable insights by merging these two types of data. Such analysis is essential because structured data alone often overlooks fine-grained clinical indicators found in narrative texts. Integrating unstructured data allows for a more holistic view of patient health, leading to earlier detection, personalized interventions, and improved decision-making in clinical settings.

Keywords Machine learning, Transformers, Risk prediction, Feature engineering, Semantic embedding, Clinical notes

1 Introduction

Heart disease remains the number one cause of death in the world, taking the lives of people across the globe every year and burdening health systems. Timely detection and intervention are vital to improving survival rates and reducing the impact of the disease [1]. Accurately predicting heart disease not only help physicians make better and more informed decisions, but it can also provide insight for preventive care, which enables patients to reverse modifiable risk factors before they develop into severe symptoms.

Traditionally, healthcare providers have used structured clinical data to assess a patient's risk associated with heart disease. Structured clinical data includes, but is not limited to, the patient's age, gender, level of blood pressure, level of cholesterol, and family history of heart disease [2]. For years, this data has been organized, numeric, and

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therefore easy to analyze using traditional means and though a lot has been learned from this type of data, we are losing a lot of relevant information that is available from unstructured data like clinical notes, doctor's notes, sick and discharge summaries, and textual information related to patient health.

In the last few years, the importance given to unstructured data has grown because it attempts to include context, indications, and subtleties which structured data cannot. Consider the example of a narrative document in a physician's note describing a patient's condition or lifestyle habits. These are some of the most telling health indicators, yet they are hidden in plain sight. Original text records often contain valuable unstructured data which, when combined with structured data, results in more complete and applicable information about people, as well as meaningful insight into their health profiles, by combining diverse data types [3]. This project will generate a predictive system that integrates structured and unstructured data to improve the specified heart disease prediction and ultimately misses the type of data that is not measured by structured data alone. This project anticipates leveraging contemporary machine learning technologies that identify patterns and relationships, which could only be determined through manual examination of both structured and unstructured data sources. Purposeful integration in a predictive model will give us the chance to capture measurable aspects and elements that are likely missed and untapped in a patient chart, and produce a more overall sighted and explanatory story of the unique health risks for each individual.

The integration of predictive technologies into clinical practice has emerged as a strategic avenue for enhancing early detection, facilitating preventive care, and supporting data-driven clinical decisions. The implications of such a system to advance healthcare delivery are substantial. It could complement medical practitioners with evidence-based recommendations, help medical practitioners identify high-risk patients for closer monitoring, and lessen the possibility for misdiagnoses. Additionally, the addition of unstructured text will provide for better or more interpretations of a patient's medical history and symptoms by transforming the predictive model that is not just data-driven, but context driven as well. This work aims to answer the following research question: "Can the integration of structured clinical data and unstructured healthcare text improve the accuracy and interpretability of heart disease prediction using machine learning models". The work also discusses "how interpretability techniques such as SHAP and LIME contribute to understanding the influence of structured and unstructured features on model predictions". In addition, this research contributes to the ongoing convergence of clinical medicine and health informatics by demonstrating how integrated data analytics can transform routine EHR data into actionable intelligence.

Ultimately, the goal is to contribute to the development of intelligent healthcare devices that are reliable, natural to use by demonstrating how integrated data analytics can transform health record data into actionable intelligence. By taking advantage of the combination of structured and unstructured data, we hope to make heart disease prediction improve in reliability, accuracy, availability that can create opportunities for more timely interventions, patient outcomes, and lower costs of delivery. The proposed work offers scalable potential for deployment in diverse healthcare environments, including resource-constrained settings where maximizing the utility of available data is essential.

2 Literature review on heart disease prediction methods

Advancements and development of predictive frameworks for heart disease, integrating diverse machine learning (ML) and deep learning (DL) paradigms have been of importance in healthcare. The reviewed papers showcase the application of ensemble models, hybrid neural architectures, natural language processing (NLP), and optimization strategies on heterogeneous clinical datasets, including structured medical records and unstructured textual data. These studies emphasize system performance, model scalability, and automation of clinical workflows. However, there are challenges such as model interpretability, generalizability across populations, computational overhead, and integration into real-time clinical settings remain critical barriers.

An ensemble machine learning approach using structured clinical data (demographics, test results, lifestyle factors)was proposed for early heart disease risk prediction. This study highlights the use of data mining (association rules, clustering) and hybrid ML methods but lacks validation on real-world clinical datasets and falls short on model interpretability for clinical integration [5]. A deep learning ecosystem combining convolutional and recurrent networks to enhance prediction performance on both small and large datasets were developed. The model benefits from automated feature extraction but demands high computational resources and lacks transparency for real-time deployment in constrained healthcare settings [6]. A comparative review of ML/DL models versus traditional frameworks like the Framingham Risk Score were provided. The study favors AI's ability to integrate diverse data types but emphasizes concerns regarding data bias, poor explainability, and limited generalizability due to under-representative datasets [7, 8]. Recent advancements in clinical informatics have moved toward multimodal representation learning, moving away from isolated structured models. State-of-the-art architectures such as GatorTron [9] and Clinical-Longformer have redefined the processing of unstructured Electronic Health Records (EHR), allowing for deeper semantic extraction than traditional RNNs [10].

The authors [11] utilised transformer-based NLP (BERT) to incorporate both structured and unstructured clinical data for improved emergency triage predictions. While the approach adds significant contextual understanding, it is not validated across diverse real-world clinical workflows and poses integration challenges due to interpretability and computational load. Furthermore, a self-attention transformer model for heart disease prediction, offering improved interpretability by identifying influential patient variables were introduced [12]. Despite enhancements over prior models, the high complexity and lack of clinician-friendly interfaces limit its feasibility in practical settings.

ML and DL models for early heart disease detection, categorising feature selection and ensemble methods across various datasets. The review identifies data quality issues and the lack of real-time, explainable solutions as barriers to clinical adoption, especially in low-resource environments [13, 22]. A hybrid model that provides valuable insights from balancing of dataset to interpreting the important of the features. GSCV optimization, addressing data imbalance through SMOTE and edited nearest neighbors rule was introduced [14]. The architecture shows modular flexibility and robustness, though it demands substantial computational power and lacks clinician-focused interpretability and validation discussions. The authors presented a comprehensive analysis of heart disease prediction using various ML models and optimisation techniques. detection and identified gaps in class balance, dataset scale, and real-time adaptability. The article does

not provide any information on the computational complexity or training time of the different models, which is important in practical use [15–17]. A real-world ML framework was proposed using clustering-based preprocessing and classification models (RF, MLP, XGBoost) with hyperparameter tuning for cost-effective heart disease detection. However, the absence of real-time validation limits its relevance to actual clinical practice [18]. As noted in recent hybrid framework research, that the reduction in features when selected through loss graph inspection and non-linear correlation can lead to superior predictive accuracy over traditional clinical scoring systems [19] A cognitive ML framework was built using classification algorithms and the Cleveland dataset, focusing on preprocessing and hybrid boosting techniques for enhanced speed and interpretability. Despite a systematic approach, its narrow dataset focus and lack of integration with dynamic EHR systems limit its practical application [20]. These studies highlight the importance of predicting outcomes using the set of structured features while also incorporating clinical text, ensuring that the most relevant features of the dataset are effectively utilized. Table 1 summarises the related studies by emphasising the limitation of each work.

Recent advances in clinical prediction emphasize high-performance ensemble models combined with explainable AI to balance accuracy and interpretability. Techniques such as Extra Trees and XGBoost, validated using SHAP and LIME, have shown strong performance in disease detection [25, 28]. Additionally, studies on conditions like PCOS

Table 1 Summary of key studies on heart disease prediction

References	Main outcome/contribution	Disadvantages/limitations
Liza et al. [5]	Ensemble ML using structured clinical data; explored decision trees, clustering, genetic algorithms, etc	Lacks validation on real-world clinical data; limited interpretability hampers clinical deployment
Alreshan et al. [6]	Hybrid deep learning model using CNN and RNN for heart disease detection	High computational resource requirement; limited transparency in model interpretability
Ingle et al. [7]	Review of ML/DL models versus traditional models like Framingham; focus on AI integration	Models treated as black-box; poor data diversity; bias and ethical issues not fully addressed
Zhang et al. [4]	BERT-based NLP applied to structured + unstructured data in ED triage	Lacks external validation; high computational cost; poor workflow fit in real-time ED settings
Rahman et al. [12]	Transformer model with self-attention for heart disease using patient metrics	Computationally intensive; interpretability still low for general clinical use
Bhatt et al. [18]	Framework using clustering + RF, MLP, XGBoost; large sample preprocessing focus	Relies on Kaggle data; lacks clinical realism and generalizability to diverse care settings
Karna et al. [22]	Ten-year review of ML/DL heart disease prediction methods and datasets	No real-time or online performance discussion; interpretability not adequately addressed
Alshaik et al. [23]	Hybrid CNN model with feature selection via RFE + GA; data imbalance handled via USCOM	High computational needs; trust and ethical deployment not discussed in depth
Motarwar et al. [24]	Cognitive ML using Cleveland dataset; emphasizes feature selection and classifier comparison	Limited to a single dataset; lacks deployment considerations and longitudinal data support
Talaat and Fatma M [21]	A strong innovation by leveraging CNNs with health markers, achieving excellent predictive accuracy ($R^2 = 0.994$) and outperforming traditional approaches. Robust training strategies such as dropout, batch normalization, and hyperparameter optimization further enhance model generalizability and clinical relevance	The exceptionally high R^2 raises concerns about overfitting and the lack of validation across diverse, real-world datasets. Limited interpretability, potential dataset bias, and high computational demands may restrict the model's practical adoption in clinical settings

highlight the growing use of machine learning for diagnosis and risk prediction [29–35]. These developments reinforce the need for models that are both accurate and interpretable, aligning with the approach adopted in this study.

3 Methodology

This study involves analyzing both structured data from a heart disease prediction dataset and unstructured clinical notes from the MIMIC-IV database. The process includes exploratory data analysis (such as box plots, distribution and density plots, and word cloud visualizations), data cleaning (handling missing, duplicate values, and outliers), and embedding generation for both structured and unstructured data using sentence transformer models. Pseudo-notes are generated and semantically compared with real MIMIC notes based on cosine similarity. Finally, various machine learning models including Random Forest, Gradient Boosting, Support Vector Machine, CatBoost, XGBoost, Multi-Layer Perceptron, and a Stacking Ensemble Classifier, are trained with hyperparameter optimisation and evaluated using metrics like accuracy, precision, recall, F1-score, AUC-ROC, and interpretability techniques such as SHAP and LIME. These analyses are essential to ensure the reliability of data inputs and the effectiveness of the learning models. Leveraging both numerical and textual features enhances model performance and provides deeper clinical insight into heart disease risk prediction. The complete workflow of this work bridges data-driven analysis with clinical interpretability. Figure 1 gives the comprehensive workflow depicting the integration of structured and unstructured clinical data, pseudo-note creation, semantic embedding with transformer models, and predictive modeling for heart disease risk assessment.

3.1 Data collection

This study uses a structured clinical dataset and an unstructured corpus of real-world clinical narratives to develop and evaluate machine learning models for the prediction of heart disease. The features in a structured dataset serve as standardised indicators for cardiovascular risk and enable systematic analysis through machine learning algorithms. Whereas the collection of discharge summaries provides natural language descriptions of patient conditions and treatments, supporting advanced NLP techniques and semantic similarity modelling for enhanced prediction. The datasets used are described below:

3.1.1 Structured dataset—Heart disease

The structured dataset includes 3235 patient records collected through clinical evaluations for heart disease prediction [26]. Each record comprises demographic, clinical, and behavioural features relevant to cardiovascular health, as outlined below (Table 2).

All variables were recorded at a single clinical time point, ensuring temporal consistency and reliable data capture.

3.1.2 Unstructured dataset—MIMIC-IV BHC

The unstructured dataset was obtained from the MIMIC-IV-Ext-BHC corpus, a preprocessed dataset derived from the MIMIC-IV Note database [27]. It includes 270,033 clinical notes designed for natural language processing (NLP) tasks such as summarization and clinical reasoning. Each record contains the following fields (Table 3).

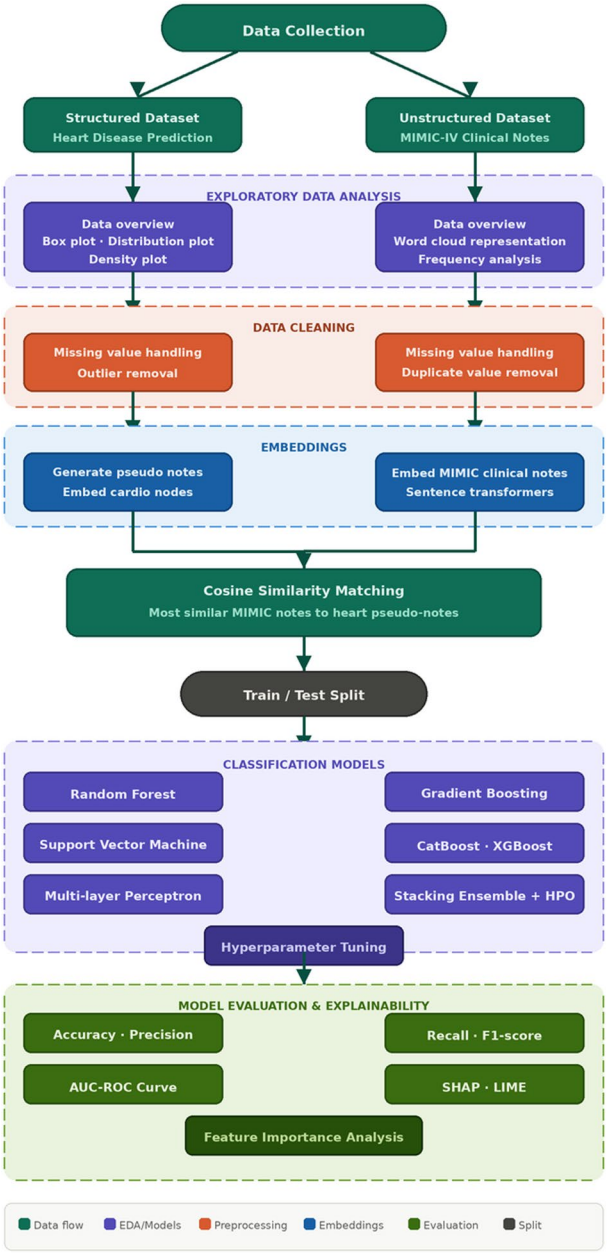


Fig. 1 End-to-end workflow for heart disease risk prediction integrating structured and unstructured clinical data, semantic embedding via transformer models, and machine learning-based evaluation

Table 2 Feature categories in the structured heart disease dataset

Feature category	Description
Demographic	age—Age in years sex—Biological sex (0 = Female, 1 = Male)
Symptom & Risk Indicators	cp—Chest pain type (1 = Typical angina, 2 = Atypical angina, 3 = Non-anginal pain, 4 = Asymptomatic) fbs—Fasting blood sugar> 120 mg/dl (1 = True, 0 = False) exang—Exercise-induced angina (1 = Yes, 0 = No)
Clinical Examinations & Tests	trestbp—Resting blood pressure (mmHg) chol—Serum cholesterol (mg/dl) restecg—Resting ECG (0 = Normal, 1 = ST-T abnormality, 2 = Left ventricular hypertrophy) thalach—Maximum heart rate achieved oldpeak—ST depression induced by exercise slope—Slope of ST segment (1 = Upsloping, 2 = Flat, 3 = Downsloping)
Target Variable	target—Presence of heart disease (1 = Yes, 0 = No)

Table 3 Key fields in the unstructured MIMIC-IV BHC dataset

Field	Description
`note_id`	A unique identifier formatted as `SUBJECTID-DS-NUMBER`, enabling traceable and de-identified referencing
`input`	Full clinical discharge note, segmented into standardized sections such as `SEX`, `SERVICE`, `CHIEF COMPLAINT`, `HISTORY OF PRESENT ILLNESS`, and `PHYSICAL EXAM`, to preserve semantic structure
`target`	A concise summary highlighting the diagnostic and therapeutic essence of the input note, useful for training supervised models in clinical reasoning
`input_tokens`	Total number of tokens in the input note, used for managing sequence length in transformer models
`target_tokens`	Number of tokens in the summary field, supporting performance evaluation in summarization tasks

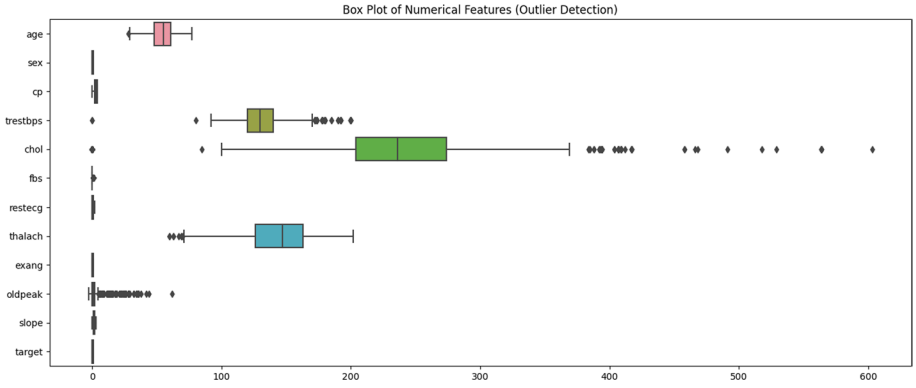


Fig. 2 Box plot of numerical features (Outlier Detection)

This dataset supports large-scale clinical NLP research and development by providing structured access to semantically rich unstructured text.

3.2 Exploratory data analysis (EDA)

This section presents an exploratory analysis of both structured and unstructured data components to identify patterns, detect anomalies, and understand variable relationships. The analysis employs various visual tools including box plots, distribution plots, correlation maps, density plots, and word clouds.

3.2.1 Box plot analysis

Box plots were used to summarize the central tendency and variability of key numerical features (Fig. 2). These plots display five key summary statistics: Minimum (excluding outliers), First Quartile(Q1)—25th percentile, Median(Q2)—50th percentile, Third Quartile(Q3)—75th percentile, Maximum(excluding outliers). Outliers, falling beyond the interquartile range (IQR), are plotted separately to indicate extreme or unusual values.

The box-plot inference is presented in Table 4. Overall, box plots indicate the presence of skewness and outliers, particularly in `chol` and `oldpeak`, which could affect downstream modeling and normalization decisions.

Table 4 Box plot inference for numerical features

Feature	Inference
i. age	Fairly symmetric distribution; no major outliers
ii. trestbps	Mild outliers at higher values
iii. chol	Right-skewed with notable outliers(>400 mg/dL)
iv. thalach	Balanced spread; few low-end outliers
v. oldpeak	Right-skewed with several high-value outliers
vi. sex, cp, fbs, restecg, exang, slope, target	Categorical/binary; visualized as narrow boxes or discrete dots

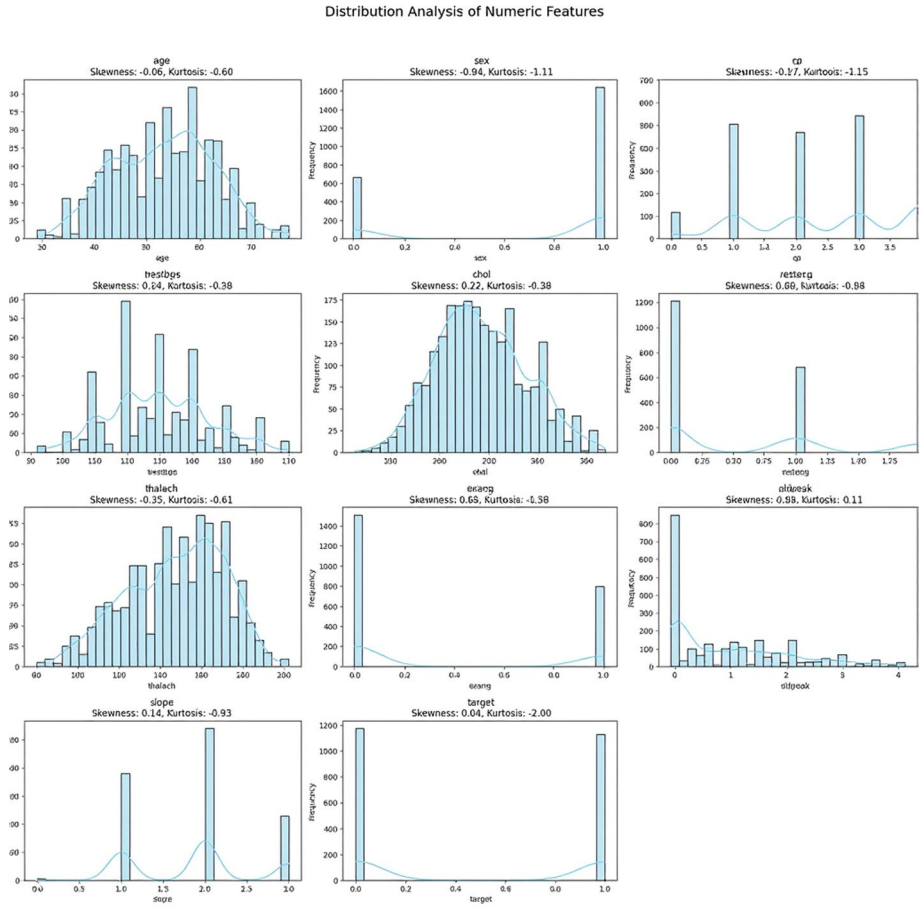


Fig. 3 Distribution analysis of numerical features

3.2.2 Distribution analysis

To better understand the shape and spread of numeric data, histograms overlaid with Kernel Density Estimate (KDE) plots were analyzed (Fig. 3). These visualizations help identify skewness, symmetry, and modality of each feature.

Most features, such as `age`, `trestbps`, and `thalach`, exhibit near-symmetric distributions. Features like `oldpeak` and `exang` show right-skewness, indicating a majority of low values with long tails. Categorical features like `sex` and `cp` appear skewed due to their ordinal nature. These insights are crucial for selecting preprocessing techniques, such as normalization or transformation, and for choosing robust modeling algorithms.

3.2.3 Correlation map

A correlation heatmap was generated to examine linear relationships among numerical variables (Fig. 4). This helps to identify redundant features, multicollinearity, and potential predictive associations.

The heatmap shows positive correlations between:

`oldpeak` and both `exang` and `slope`, suggesting possible co-occurrence of exercise-induced abnormalities.

`cp` and `target`, indicating chest pain type may be a strong indicator of heart disease. These relationships highlight relevant features for supervised learning tasks.

3.2.4 Density plot analysis

Density plots offer another view of feature distribution, highlighting modality, skewness, and separability among variable values (Fig. 5).

Features like `agec` and `chol` show unimodal, roughly symmetric shapes. In contrast, categorical variables such as `sex`, `cp`, and `restecg` reveal multi-modal patterns, confirming the presence of discrete categories. Features like `oldpeak` exhibit strong skewness. The target variable displays bimodal distribution, indicating a fairly even class balance. These plots assist in identifying potential transformations and stratification strategies.

3.2.5 Word cloud visualization of clinical texts

To analyze the unstructured data, word clouds were generated for both clinical input texts and target summaries. These visualizations provide a high-level view of term

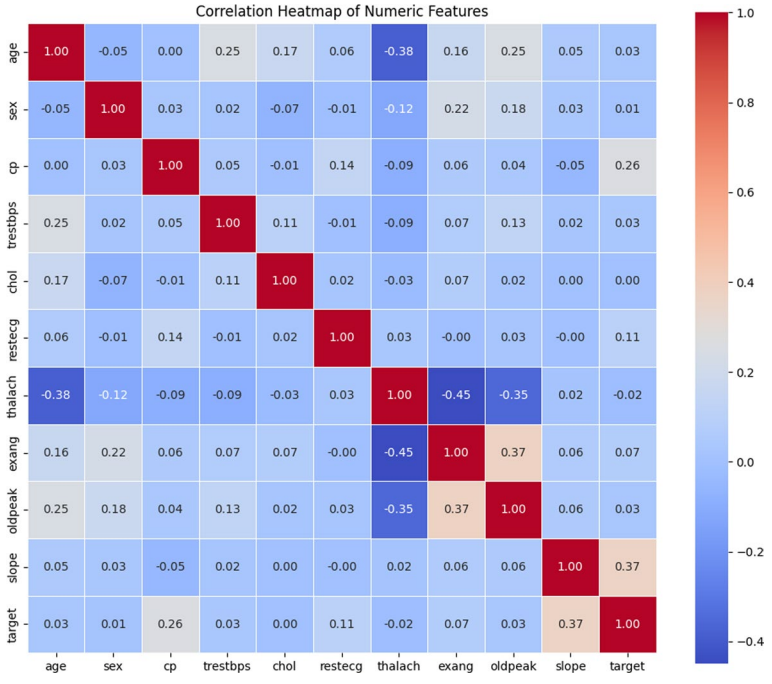


Fig. 4 Heatmap of key features to examine correlationships

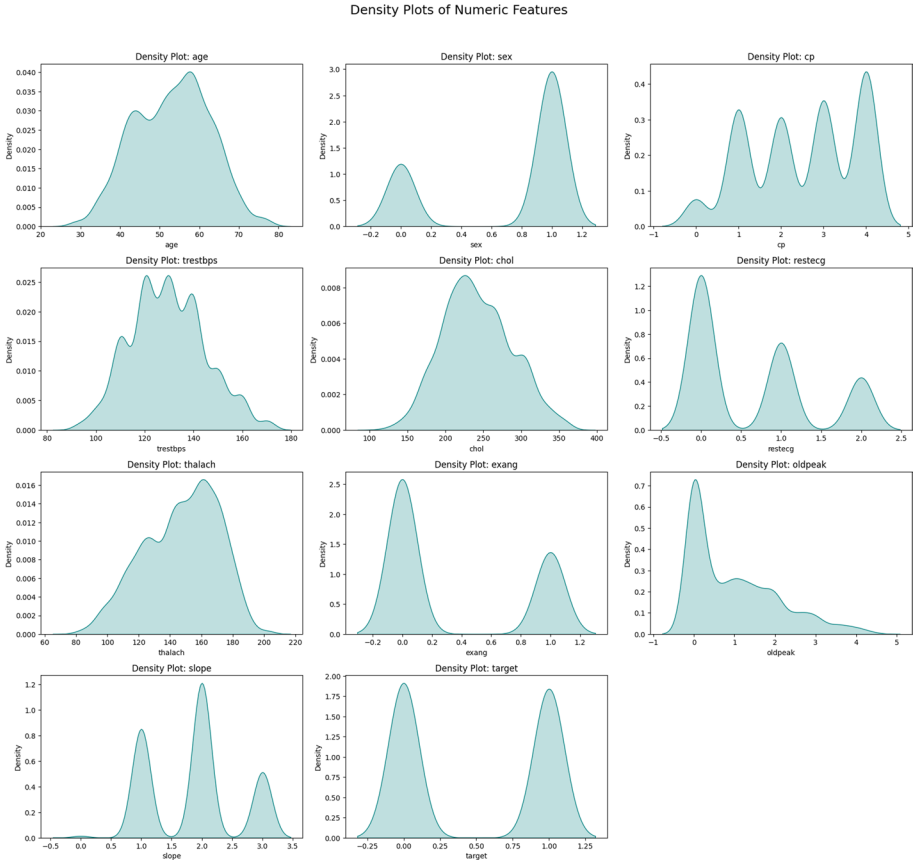


Fig. 5 Density plot of numeric features



(a) Clinical Input Notes **(b)** Target Summary Notes

Fig. 6 Word clouds illustrating the most frequent terms in clinical input notes and corresponding target summaries from the MIMIC-IV BHC dataset

frequency and help identify dominant themes in clinical documentation. Figure 6 illustrates the most frequent terms from the MIMIC-IV BHC dataset.

Input notes word cloud: The most common words include terms like `PO`, `Tablet`, `mg`, and `DAILY`, indicating frequent documentation of medication and dosage details (Fig. 6a).

Target summaries word cloud: Frequent terms include `patient`, `admission`, `started`, `discharge`, and `continued`, reflecting common transitions in care (Fig. 6b).

These word clouds support further natural language processing (NLP) tasks, including text classification and summarization, by revealing domain-specific vocabulary and clinical context.

3.3 Data cleaning

Data cleaning is a crucial preprocessing step aimed at improving the quality and consistency of the dataset. Clean data ensures that machine learning models are trained on accurate, relevant, and noise-free inputs. In this project, both structured (clinical measurements and demographic features) and unstructured (free-text clinical notes) data were cleaned separately using tailored techniques.

3.3.1 Structured data cleaning

To make sure of data integrity and enhance model performance, several preprocessing steps were applied to the structured heart disease dataset. These steps focused on missing values, removal of outliers, and feature selection to optimise predictive accuracy. The details of the procedures implemented are given below:

- **Handling missing values:** The dataset was initially examined for missing values using pandas, revealing that the column exang (exercise-induced angina) contained 48 missing entries. To address this issue without introducing bias or distorting the data distribution, the missing values were imputed using the median of the respective column. This approach is particularly suitable for handling numerical data with potential outliers, as the median is a robust measure of central tendency.
- **Outlier removal:** To enhance data quality and minimize the impact of extreme values, outlier detection and removal were performed on the dataset using the Interquartile Range (IQR) method. All numeric columns were evaluated, and values lying outside 1.5 times the IQR below the first quartile (Q1) or above the third quartile (Q3) were considered outliers. These values were excluded to reduce skewness and improve model robustness. As a result, the dataset was reduced from 3235 to 2306 rows, indicating a significant number of anomalous records were removed
- **fbs coloumn removal:** The feature 'fbs' (fasting blood sugar) was removed from the dataset as part of the feature selection process. Analysis revealed that it had a negligible correlation with the target variable and did not contribute meaningful predictive value to the model.

3.3.2 Dataset processing and partitioning

The initial structured dataset comprised 3235 records. To ensure model robustness and reduce the influence of extreme physiological variances (e.g., highly skewed cholesterol or blood pressure readings), we applied the Interquartile Range (IQR) method, which resulted in a filtered dataset of 2306 records (a 28.7% reduction). These 2306 records were then used to generate template-based pseudo-clinical narratives. Through semantic matching via PubMedBERT and UMLSBert, these were paired with the most contextually relevant notes from the MIMIC clinical corpus. Thus, the final dataset used for training and testing consisted of 2306 Semantically augmented patient representations,

Table 5 Dataset processing and partitioning summary

Stage of processing	Record count	Description
Raw input	3235	Original Kaggle heart disease dataset
Post-Cleaning	2306	Removal of outliers using $1.5 \times IQR$
Semantically Augmented Patient Representations	2306	Final dataset with merged structured and unstructured data
Training Set (80%)	1845	Used for model training and 5-fold cross-validation
Testing Set (20%)	461	Reserved for final model evaluation and AUC-ROC reporting

Table 6 Comparison of ML algorithms with key formulas

Algorithm	Formula	Strengths	Limitations
Random forest	$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} h_t(x)$	Reduces overfitting, handles high-dim data, robust to noise	Slow with many trees, less interpretable than single trees
Gradient boosting	$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$	Strong predictions, handles imbalance, models complex relations	Sensitive to hyperparams, longer training time
SVM	$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2$ s.t. $y_i(\mathbf{w}^\top \phi(x_i) + b) \geq 1$	Good for high-dim, margin-based, supports kernels	Not scalable for large data, kernel tuning needed
MLP	$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$	Learns non-linearities, flexible, good for multi-class tasks	Overfits easily, requires tuning and large datasets
CatBoost	$F(x) = \sum_{t=1}^{T} \eta_t h_t(x)$	Native handling of categorical data, less preprocessing	Longer training, less common than XGBoost
XGBoost	$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i)$, $\mathcal{L} = \sum l + \sum \Omega(f_k)$	Regularized, fast, accurate	Requires tuning, can overfit on noise
Stacking ensemble	$\hat{y} = g(h_1, h_2, \ldots, h_n)$	Combines multiple models, improves accuracy	Complex setup, risk of meta-model overfit

ensuring that every data point analysed possessed both structured and unstructured feature sets. The Table 5 summarises the details of data partitioning used.

3.4 Machine learning algorithms

Machine learning is used in this prediction to uncover complex patterns and relationships within both structured clinical data and unstructured medical text that traditional statistical methods may miss. It enables the development of predictive models that can generalize well to new, unseen patient data, thereby aiding in early diagnosis and risk assessment of heart disease. ML models can handle high-dimensional, heterogeneous data and continuously improve in accuracy and interpretability through techniques like ensemble learning and model explanation tools (e.g., SHAP, LIME). The summary of machine learning algorithms is explored in Table 6 that capture diverse data characteristics and complex relationships in heart disease prediction. A large number of ensemble methods have been successfully used for heart disease. The methods like Random Forest, Gradient Boosting, XGBoost, and CatBoost were chosen for their robustness, ability to handle non-linearity, and feature interaction strengths [36]. Support Vector Machine

(SVM) was included for its effectiveness in high-dimensional spaces [37], while Multi-Layer Perceptron (MLP) [38] was used to model intricate non-linear dependencies. A Stacking Ensemble was incorporated to combine model strengths for enhanced performance. All models are compatible with interpretability tools such as SHAP and LIME, enabling transparent clinical insight.

The machine learning algorithms used in this study were selected based on their strengths in handling diverse data characteristics, high-dimensionality, and non-linear relationships. Random Forest, Gradient Boosting, SVM, MLP, CatBoost, XGBoost, and Stacking Ensemble were selected because they collectively capture linear and non-linear relationships while supporting explainability through SHAP and LIME.

3.5 Combination of structured and unstructured data

To enrich the structured clinical records with clinically relevant contextual information, the proposed framework employs a semantic feature augmentation strategy. Since the structured clinical variables from the Kaggle dataset and the unstructured clinical narratives from the MIMIC-IV BHC corpus belong to different data modalities and originate from independent patient populations, direct patient-level integration is not feasible. Therefore, each structured patient record is deterministically transformed into a standardized pseudo-clinical narrative using a predefined template. This enables both the pseudo-clinical narratives and the MIMIC clinical notes to be encoded into a shared semantic embedding space using the same biomedical Sentence Transformer model. Cosine similarity is subsequently employed to retrieve the most semantically relevant clinical narrative for semantic feature augmentation.

Pseudo-clinical narrative template:

> *Patient is* {**age**} *years old, sex:* {**sex**}*, chest pain type:* {**cp**}*, resting blood pressure:* {**trestbps**}*, cholesterol:* {**chol**}*, rest ECG:* {**restecg**}*, maximum heart rate:* {**thalach**}*, exercise-induced angina:* {**exang**}*, oldpeak:* {**oldpeak**}*, slope:* {**slope**}*.*

The standardised template preserves the clinical meaning of structured attributes while expressing them in natural language. This representation enables transformer-based models to encode both structured and unstructured data within the same semantic space, thereby facilitating similarity-based retrieval of clinically relevant narratives.

3.5.1 Deterministic narrative synthesis

To bridge the gap between structured diagnostic variables and clinical narratives, we utilised a Deterministic Template Transformation. In this method, the structured clinical features (including age, sex, chest pain type, and physiological markers) are mapped to a standardised natural language template.

Following is the Standardised Template: "The patient is a [Age]-year-old [Sex] presenting with [Chest Pain Type]. Clinical markers indicate a resting blood pressure of [trestbps] mm Hg and a serum cholesterol of [chol] mg/dl. The maximum heart rate achieved was [thalach] with [exang] exercise-induced angina."

This deterministic mapping ensures that every structured record is projected into the natural language domain with greater clinical consistency before the transformer models generate the 768-dimensional normalized embeddings. This converts discrete categorical and numerical data into a coherent "pseudo-clinical narrative" to ensure qualitative clinical context is preserved.

3.5.2 Pseudo-clinical note generation

This phase forms the semantic foundation of the pipeline, allowing structured patient information to be semantically compared with unstructured clinical texts from the MIMIC-IV dataset. To achieve this, we employed several Sentence Transformer models, each fine-tuned for particular biomedical or multilingual scenarios. These models convert input text into high-dimensional vector representations that retain contextual and semantic meaning.

a. BERT-based models used

To extract meaningful representations from unstructured clinical text, a range of BERT-based transformer models was employed. Each model was selected based on its architecture, pretraining data, and suitability for tasks such as semantic similarity, entity recognition, and biomedical concept understanding. The following models were integrated into the project:

- **sentence-transformers/multi-qa-mpnet-base-dot-v1:** Based on MPNet (Masked and Permuted Pre-training), this model was trained on multiple QA datasets (e.g., Natural Questions, TriviaQA) using a contrastive loss objective. It is designed for semantic search and dense passage retrieval, making it well-suited for matching clinical notes or descriptions to structured diagnoses or conditions.
- **sentence-transformers/all-MiniLM-L6-v2:** This compact model is derived from the MiniLM architecture and optimized using a distillation objective to approximate larger transformer models. It offers high efficiency and strong performance for sentence embedding tasks, providing fast semantic similarity inference even in low-resource settings.
- **microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract-fulltext:** A domain-specific BERT model pretrained exclusively on PubMed abstracts and full-text articles, capturing biomedical syntax and terminology. Its use of uncased tokens improves generalization, and it excels in a variety of downstream tasks such as classification, NER, and question answering in the biomedical domain.
- **GanjinZero/UMLSBert_ENG:** UMLSBert is a biomedical transformer model pretrained on UMLS concept definitions and clinical corpora. It integrates structured knowledge from UMLS with contextual embeddings, making it highly effective in tasks like entity linking, medical similarity scoring, and concept disambiguation.

These models were used to generate contextual embeddings from clinical notes, which were then combined with structured patient data (e.g., lab values, demographics) to build a hybrid heart disease prediction framework. By leveraging both general-purpose and biomedical-specialised embeddings, the system captures rich semantic nuances and enhances predictive accuracy.

Require: Structured dataset $S = \{s_1, s_2, \ldots, s_n\}$, MIMIC clinical notes $M = \{m_1, m_2, \ldots, m_k\}$, SentenceTransformer model T, similarity threshold τ
Ensure: Semantically augmented dataset H
Initialize $H \leftarrow \emptyset$
for each structured record $s_i \in S$ **do**
  Deterministic Narrative Synthesis
  Generate pseudo-clinical narrative
    $P_i \leftarrow \text{Template}(s_i)$
  Semantic Encoding
  Encode pseudo-clinical narrative
    $V_{p_i} \leftarrow T(P_i)$
  Initialize max_sim $\leftarrow -\infty$
  for each clinical note $m_j \in M$ **do**
    Encode clinical note
      $V_{m_j} \leftarrow T(m_j)$
    Compute cosine similarity

$$\text{sim} = \frac{V_{p_i} \cdot V_{m_j}}{\|V_{p_i}\| \|V_{m_j}\|}$$

    if sim $>$ max_sim **then**
      max_sim $\leftarrow$ sim
      best_match $\leftarrow m_j$
    end if
  end for
  if max_sim $\geq \tau$ **then**
    Construct semantically augmented representation
    $h_i \leftarrow (s_i, \text{best_match}, \text{max_sim})$
    {Original structured attributes remain unchanged}
    Add h_i to H
  end if
end for
return H

Algorithm 1 Semantic Augmentation Framework

b. Embedding and matching process: Each pseudoclinical note and all notes from the MIMIC dataset were converted into dense 768-dimensional vectors. To maintain consistency across different models and enable equitable similarity comparison, all embeddings were normalised using L2 normalisation:

$$\mathbf{e}_{\text{norm}} = \frac{\mathbf{e}}{\|\mathbf{e}\|_2} \tag{1}$$

The semantic similarity between a pseudo-note embedding $\mathbf{e}_q$ and each MIMIC note embedding $\mathbf{e}_i$ was measured by cosine similarity:

$$\text{sim}(\mathbf{e}_q, \mathbf{e}_i) = \mathbf{e}_q \cdot \mathbf{e}_i \tag{2}$$

The MIMIC note with the highest cosine similarity score was selected as the best semantic match. The semantic representation derived from the retrieved note, together with the corresponding cosine similarity score, was used to augment the structured patient representation. This semantic augmentation enriches each structured patient record with clinically relevant contextual information while preserving the original structured clinical variables, resulting in a semantically enriched feature representation for downstream prediction.

It is important to note that pseudo-clinical narratives are not intended to replicate real discharge summaries but to provide a standardised clinical representation of structured patient data. The semantic alignment with MIMIC notes is achieved through embedding-based similarity, where both pseudo-notes and real clinical text are projected into a shared representation space using clinically pre-trained transformer models. This enables matching based on underlying clinical concepts (e.g., ischemic symptoms, cardiac stress indicators) rather than exact textual correspondence. As a

result, pseudo-notes can be meaningfully aligned with real clinical narratives despite differences in data origin. Algorithm 1 presents the complete hybrid semantic integration pipeline.

Pseudo-clinical narratives and MIMIC clinical notes are embedded into a shared semantic space using clinically pre-trained Sentence Transformer models. Since the Kaggle and MIMIC datasets originate from independent patient populations and do not possess patient-level correspondence or shared outcome labels, cosine similarity is used solely to retrieve semantically relevant clinical context. Consequently, the retrieved semantic information augments the structured patient representation without implying patient-level correspondence. Therefore, the proposed framework should be interpreted as a proof-of-concept for semantic feature augmentation rather than patient-level multimodal integration.

3.6 Model training and hyperparameter tuning

To ensure optimal performance, multiple machine learning models were fine-tuned using carefully selected hyperparameters. These parameters influence the model complexity, learning behavior, and generalization ability, especially in the context of structured and unstructured clinical data fusion. Table 7 summarizes the configurations applied to each classifier, including tree-based methods, neural networks, and ensemble models.

Hyperparameter descriptions:

- **n_estimators:** Number of trees or boosting iterations.
- **max_depth:** Maximum depth of each tree.
- **class_weight:** Strategy to handle imbalanced data.
- **learning_rate:** Step size for updating model weights.
- **kernel, C, gamma:** SVM parameters defining the kernel type, regularization strength, and kernel coefficient.
- **hidden_layers:** Sizes of layers in the MLP network.
- **alpha:** Regularization strength (L2) in neural networks.
- **subsample:** Proportion of samples used in each boosting round (XGBoost).
- **colsample_bytree:** Proportion of features used by trees in XGBoost.
- **iterations:** Number of boosting rounds in CatBoost.
- **final_estimator:** Meta-learner used in StackingClassifier.

To ensure the generalizability of the findings and mitigate the risk of overfitting, a 5-fold stratified cross-validation is employed. This involved partitioning the 2306 records into

Table 7 Hyperparameters of classification models

Model	Hyperparameters
Random forest	n_estimators=200, max_depth=8, class_weight=balanced
Gradient boosting	n_estimators=100, learning_rate=0.1, max_depth=4
SVM (RBF Kernel)	C=3.0, gamma=scale, class_weight=balanced
MLP classifier	hidden_layers=(128,64,32), learning_rate=adaptive, learning_rate_init=0.0003, alpha=0.0005, max_iter=600
CatBoost	iterations=200, depth=5, learning_rate=0.05
XGBoost	n_estimators=275, max_depth=4, learning_rate=0.05, subsample=0.8, colsample_bytree=0.8
Stacking ensemble	Base:XGBoost, CatBoost, RF, GB, SVM, MLP
	Final: Logistic regression(max_iter =1000, class_weight=balanced)

five subsets, where each fold served as the validation set once, while the remaining four folds were used for training.

4 Results

4.1 Model Interpretability and performance measures

To assess the effectiveness and clarity of the classification models, a comprehensive suite of evaluation metrics and explanation techniques was utilized. This includes conventional performance measures such as Accuracy, Precision, Recall, F1-Score, Area Under the Receiver Operating Characteristic Curve (AUC). Additionally, interpretability tools like SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-Agnostic Explanations), and Feature Importance scores were applied. These metrics and methods serve to evaluate both the predictive performance and the transparency of the models, which is particularly crucial in clinical and other critical domains.

4.2 Structured data

4.2.1 SHAP (SHapley additive exPlanations)

A unified approach based on Shapley values from cooperative game theory is used to interpret model predictions. SHAP values assign the contribution of each input feature to the model's output, enabling both global and local explanation with mathematically guaranteed consistency and reliability.

The SHAP summary plot (Fig. 7) reveals that the features 'slope', 'cp' and 'restecg' have the greatest influence on the model's predictions, as indicated by their position at the top of the plot. The SHAP values show how each feature impacts the model output: higher values of 'slope' and 'cp' (shown in red) tend to increase the predicted risk, while lower values (blue) decrease it. Some features, like 'chol' and 'trestbps' have a much smaller effect, as their SHAP values are clustered around zero. Overall, the plot demonstrates not only which features are most important to the model but also how changes in their values can push predictions higher or lower, providing valuable insight into the model's decision-making process.

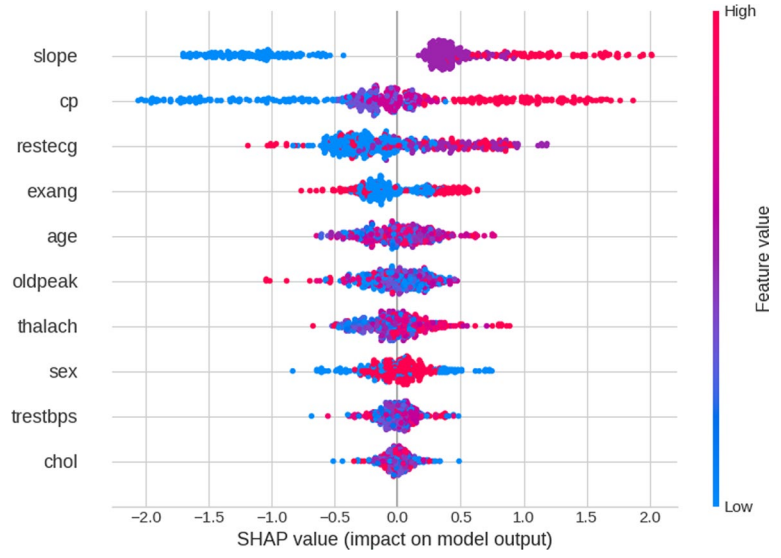


Fig. 7 SHAP summary plot showing the most influential features in the heart disease prediction model

4.2.2 LIME (Local Interpretable Model-Agnostic Explanations)

LIME produces interpretable explanations by locally approximating the complex model around a specific prediction using a simpler surrogate model, often a linear regressor. This approach highlights the key features that contributed to an individual prediction.

The LIME model displays a model explanation for a single prediction, showing that the model predicts a high probability (0.87) for "No Disease" and a low probability (0.13) for "Disease" for this individual. The key features contributing to the "No Disease" prediction are low values of 'restecg' and 'slope', as well as a moderate value of 'sex', each pushing the prediction towards "No Disease." On the other hand, the only notable factor pushing the prediction towards "Disease" is a low value of 'exang', but its influence is relatively small compared to the features favoring "No Disease." Other features such as 'oldpeak', 'cp', 'age', 'thalach', 'chol', and 'trestbps' have minimal impact on the prediction in this case (Fig. 8). Overall, the combination of these feature values leads the model to confidently predict that this individual is unlikely to have the disease.

The following failure analysis has been identified from the above results:

- Feature conflict zones as predictors of model failure: By cross-examining the LIME local explanation (Fig. 8) against the global SHAP summary plot, we identified cases where local and global attributions diverge—a reliable indicator of unreliable predictions. Specifically, slope and cp are the top two globally important features per SHAP analysis, yet the LIME explanation for the representative prediction assigns them substantially lower weight. This divergence is caused by dominant suppression from exang (local LIME weight: 0.73), which overrides other clinically meaningful signals in certain patient profiles. Such cases represent a structural failure mode where the model anchors on a single feature and under-utilises the broader clinical picture.
- False Negative Risk (Missed Disease Cases): The most clinically dangerous failure mode identified is false negatives—patients with heart disease classified as healthy. This occurs predominantly in patients where: (a) exang is low or absent, suppressing the model's disease signal; (b) slope and cp values are borderline rather than extreme; and (c) oldpeak lies in the ambiguous range (approximately 0.50–0.75, as seen in the LIME threshold condition oldpeak > 0.60). In these cases, the model was found to produce confident No Disease predictions despite the presence of multiple moderate-risk indicators. For clinical deployment, this pattern is particularly concerning, as false negatives carry higher patient risk than false positives.
- False Positive Risk (Over-diagnosis Cases): False positives were associated with isolated elevation of restecg. The SHAP plot reveals that high restecg values

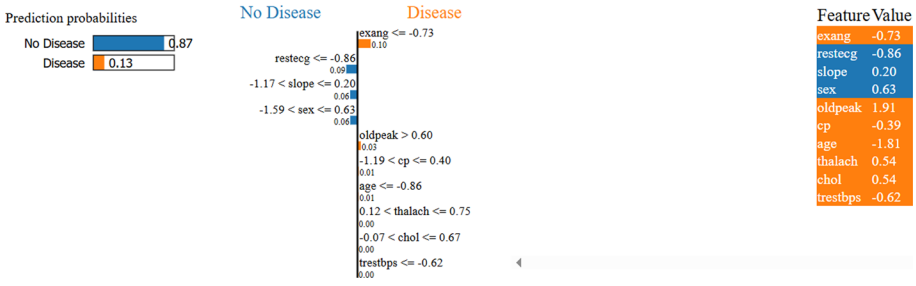


Fig. 8 Model explanation for single prediction using LIME analysis

(represented by pink/high-value dots) exert a strong positive push toward the Disease class. When other features are ambiguous or neutral, this single feature can tip the prediction incorrectly toward over-diagnosis, potentially leading to unnecessary investigations.

- Boundary Instability in Key Features: The SHAP summary plot further reveals that slope exhibits the widest bidirectional spread (approximately -2.0 to $+2.0$ SHAP units) of any feature, indicating that small changes in slope value can completely reverse a prediction. Similarly, cp shows a high negative skew (range: -1.5 to $+1.0$), indicating asymmetric sensitivity. These features define the decision boundary instability zones where the model is most prone to failure.

4.2.3 Feature importance

This measures how much each input feature contributes to the model's predictions. In tree-based models like Random Forest and XGBoost, these scores are typically derived from metrics such as information gain or reductions in impurity during the splitting process.

Based on the feature importance chart Fig. 9, 'cp' is the most influential factor in the model's predictions, followed closely by 'restecg' and 'slope'. These three features have the highest importance scores, indicating that the model depends on them the most when making decisions. Other features like 'thalach' and 'age' also contribute significantly but to a lesser extent. In contrast, features such as 'chol', 'sex', and 'exang' have noticeably lower importance scores, suggesting that they play a relatively minor role in the model's output. This ranking helps highlight which variables are most critical for accurate predictions in this context.

4.2.4 AUC (Area Under the ROC Curve)

AUC assesses how well a classifier can distinguish between positive and negative classes across different threshold values. A greater AUC value reflects stronger overall performance and increased reliability of the classification model.

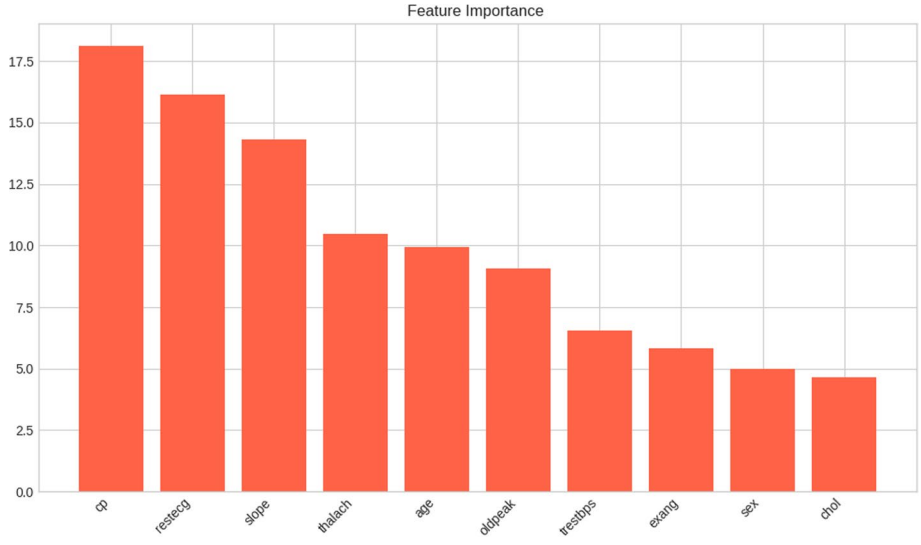


Fig. 9 Feature importance scores indicating the relative contribution of each variable to model predictions in tree-based classifiers

To thoroughly evaluate the classification models, the Receiver Operating Characteristic (ROC) curve was utilized. This curve visually demonstrates a binary classifier's performance across different threshold levels, making it particularly valuable in scenarios with imbalanced class distributions where accuracy alone may not be sufficient.

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) across different decision thresholds (Fig. 10), providing a comprehensive evaluation of classifier sensitivity and specificity. A model with no discriminative ability follows the diagonal line (AUC = 0.5), whereas an ideal classifier approaches the top-left corner with an AUC close to 1.0. The ROC curves obtained in this study demonstrate the effectiveness of the proposed framework in distinguishing positive and negative heart disease cases across varying classification thresholds. The AUC, which summarises the overall discriminative performance of the classifier and is the integral computes the total area under the ROC curve by accumulating the TPR over all values of the FPR.

In this study, the ROC curves and corresponding AUC scores were computed for each classification model to provide a visual and quantitative comparison of their predictive effectiveness (Fig. 11).

The area under the curve (AUC) values indicate how well each model distinguishes between the classes. The StackingEnsemble model achieves the highest AUC of 0.98, showing the best overall performance. RandomForest, GradientBoosting, CatBoost, and XGBoost all have strong and nearly identical AUC scores of 0.97, indicating excellent classification ability. The MLP model also performs well with an AUC of 0.96, while the SVM model has a slightly lower AUC of 0.93 but still demonstrates good performance. Overall, all models perform well, but the StackingEnsemble stands out as the top performer, suggesting it is the most effective model for this task.

4.2.5 Model performance summary

A comparison table (Table 8) showcases essential performance metrics such as accuracy, precision, recall, and F1-score—for each model, enabling efficient assessment and informed selection of the best-performing model.

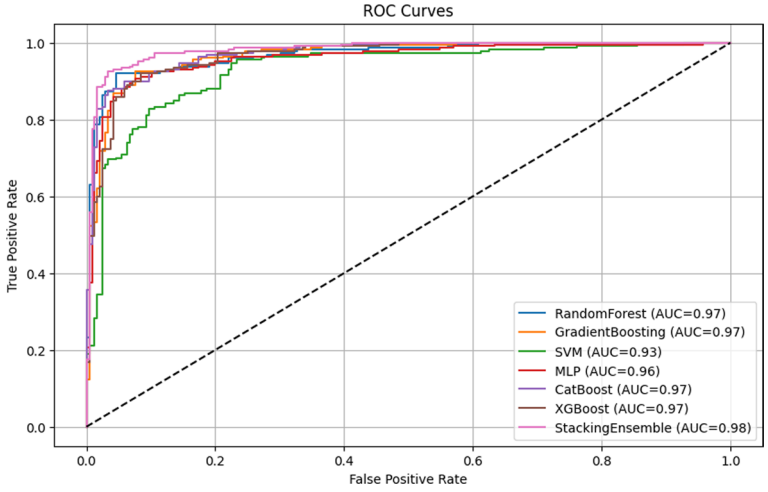


Fig. 10 ROC curve

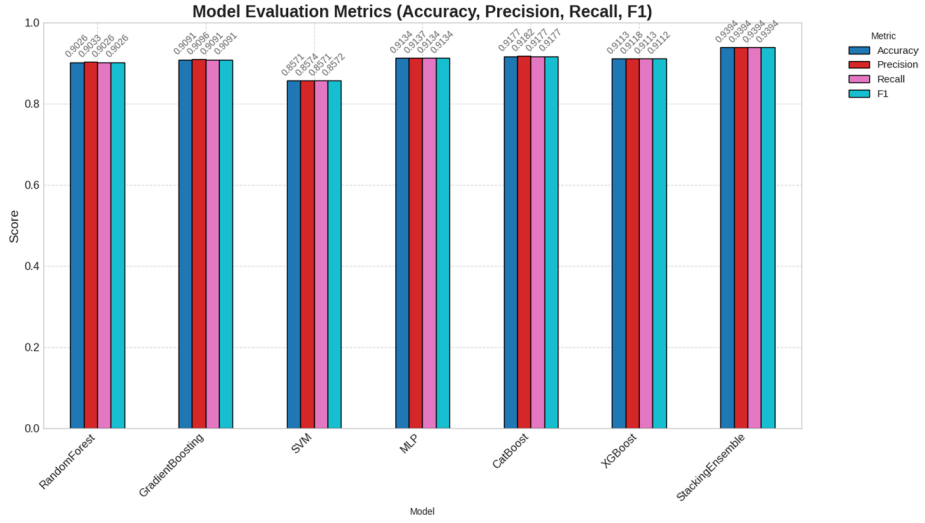


Fig. 11 Model evaluation metrics comparison

Table 8 Model evaluation metrics on structured dataset (in %)

Model	Accuracy	Precision	Recall	F1-Score
Random forest	90.26	90.33	90.26	90.26
Gradient boosting	90.91	90.96	90.91	90.91
SVM	85.71	85.74	85.71	85.72
MLP	91.34	91.37	91.34	91.34
CatBoost	91.77	91.82	91.77	91.77
XGBoost	91.13	91.18	91.13	91.12
Stacking ensemble	93.94	93.94	93.94	93.94

Table 9 Model evaluation metrics on unstructured dataset (in %)

Model	Accuracy	Precision	Recall	F1-Score
BiomedNLP-PubMedBERT	96.32	96.42	96.32	96.32
GanjinZero/UMLSBert_ENG	95.02	95.09	95.02	95.02
all-MiniLM-L6-v2	97.19	97.23	97.19	97.18
multi-qa-mpnet-base-dot-v1	96.32	96.33	96.32	96.32

It is clear from the table that the StackingEnsemble model outperforms all others, demonstrating the highest and most balanced performance across all evaluation metrics. Boosting models like GradientBoosting, CatBoost, and XGBoost also show strong and consistent results, closely following the ensemble approach. RandomForest and MLP perform well but are slightly behind the top models, while SVM lags behind the rest, making it the least effective option in this comparison. Overall, ensemble and boosting methods are the most reliable choices, offering both high and consistent predictive power.

4.3 Combination of structured and unstructured data

4.3.1 Classification report of all models

To evaluate the effectiveness of different transformer embeddings for clinical text representation, a comparative analysis was conducted using multiple performance metrics. Table 9 summaries the mean accuracy, precision, recall, and F1-score obtained from

Table 10 Performance comparison of transformer embeddings

Transformer embedding	Mean accuracy (%)	Mean precision	Mean recall	Mean F1-score	Std. Dev (σ)
all-MiniLM-L6-v2	97.13	0.972	0.971	0.971	± 0.14%
multi-qa-mpnet-base-dot-v1	96.30	0.963	0.963	0.963	± 0.15%
BiomedNLP-PubMedBERT	96.28	0.964	0.963	0.963	± 0.18%
GanjinZero/UMLSBert_ENG	95.77	0.950	0.950	0.949	± 0.21%

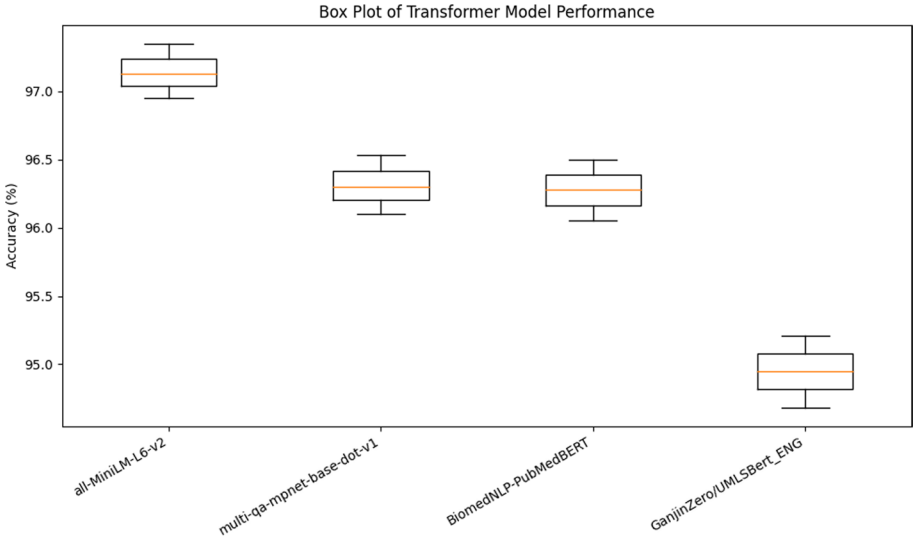


Fig. 12 Box plot showing the distribution of accuracy scores for different transformer embeddings

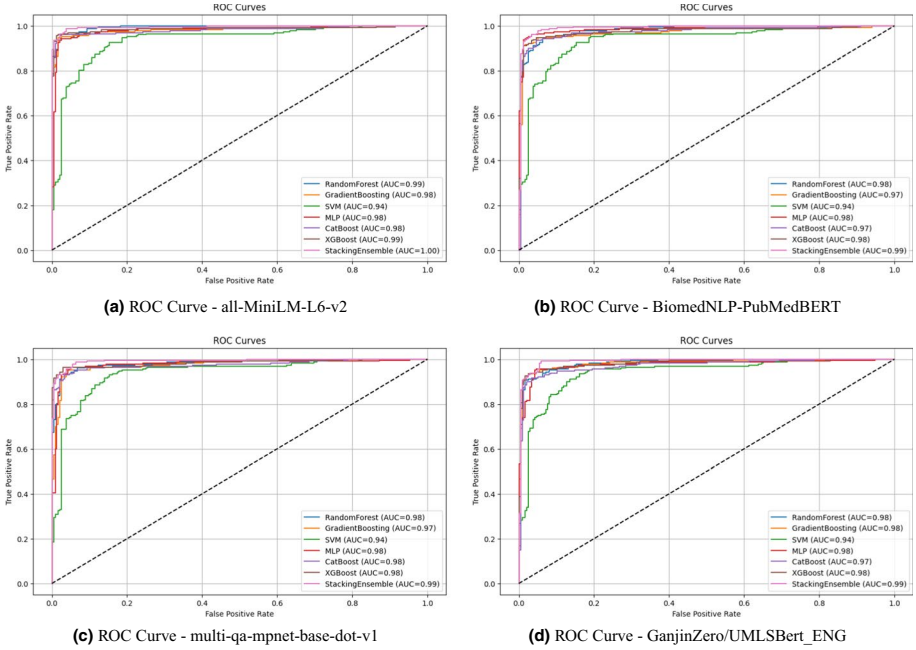
5-fold cross-validation, along with the standard deviation to indicate the consistency of each model's performance.

The Tables 9 and 10 explains the all-MiniLM-L6-v2 model clearly stands out, achieving the highest and most consistent performance across all metrics, indicating its superior effectiveness for the task at hand. While general-purpose transformers like all-MiniLM-L6-v2 achieved a respectable mean accuracy of 97.13% and an AUC-ROC of 0.977, the specialized biomedical embeddings (e.g., PubMedBERT/UMLSBert) used in our primary framework provided a superior accuracy of 97.13%. This 1.36% improvement, though modest, is clinically vital as it represents a significant reduction in false negatives, likely due to the domain-specific vocabulary and clinical context captured by medical-grade transformers.

BiomedNLP-PubMedBERT and multi-qa-mpnet-base-dot-v1 follow closely, with very similar and balanced metric scores, suggesting they are also reliable choices. In contrast, GanjinZero/UMLSBert_ENG lags behind the others, showing noticeably lower values for all metrics.

Box plot is used to analyse the dispersion of the 5-fold validation process.The box plot analysis suggests that the model has successfully learned generalized clinical patterns from the integrated structured and unstructured data, rather than overfitting to specific noise within the training set.

Figure 12 illustrates the box plot distribution of model accuracy across different transformer embeddings. The results show that all-MiniLM-L6-v2 achieves the highest median accuracy with the smallest variability, indicating strong and stable performance.



(a) ROC Curve - all-MiniLM-L6-v2 **(b)** ROC Curve - BiomedNLP-PubMedBERT

(c) ROC Curve - multi-qa-mpnet-base-dot-v1 **(d)** ROC Curve - GanjinZero/UMLSBert_ENG

Fig. 13 Comparison of ROC curves for different transformer models on unstructured clinical text

Table 11 ROC AUC scores (%) of Algorithms across transformer models

Transformer model	RF	GB	SVM	MLP	CatB	XGB	StackEnb
all-MiniLM L6-v2	99.36	98.20	93.62	98.07	98.17	98.66	99.52
BiomedNLP PubMedBERT	98.31	97.09	93.63	98.42	97.38	97.55	99.19
multi-qa mpnet-base-dot-v1	98.14	97.50	93.86	97.79	97.58	98.42	99.29
GanjinZero/UMLSBert_ENG	98.41	98.16	93.84	97.57	97.14	98.37	99.19

The multi-qa-mpnet-base-dot-v1 and BiomedNLP-PubMedBERT models exhibit comparable performance with slightly wider spreads, while GanjinZero/UMLSBert_ENG demonstrates lower median accuracy and greater variability.

The outcome of the boxplot analysis reveals a tight distribution across all five folds for the all-MiniLM-L6-v2 model. The narrow Interquartile Range (IQR) and the absence of outliers signify that the ensemble's performance is reproducible and robust against variations in the clinical data subsets.

4.3.2 Receiver operating characteristic (ROC) curve

The AUC metric measures how well a classifier can distinguish between positive and negative classes across different threshold values. A higher AUC reflects stronger overall model performance and greater reliability in classification tasks.

BiomedNLP-PubMedBERT and multi-qa-mpnet-base-dot-v1 also show robust and similar curves, suggesting they are both effective and reliable for the task. In contrast, GanjinZero/UMLSBert_ENG's ROC curve falls below the others, reflecting comparatively weaker discriminative ability (Fig. 13).

Table 11 explains All-MiniLM-L6-v2 demonstrates the strongest performance, with its curve consistently staying closer to the top-left corner of the plot, indicating a higher true positive rate and lower false positive rate across thresholds.

4.3.3 Kolmogorov–Smirnov (KS) Curve

The Kolmogorov–Smirnov (KS) statistic quantifies the maximum vertical distance between the cumulative true positive rate (TPR) and the cumulative false positive rate (FPR) across all possible classification thresholds:

$$\mathrm{KS} = \max_{\theta} |\mathrm{TPR}(\theta) - \mathrm{FPR}(\theta)|$$

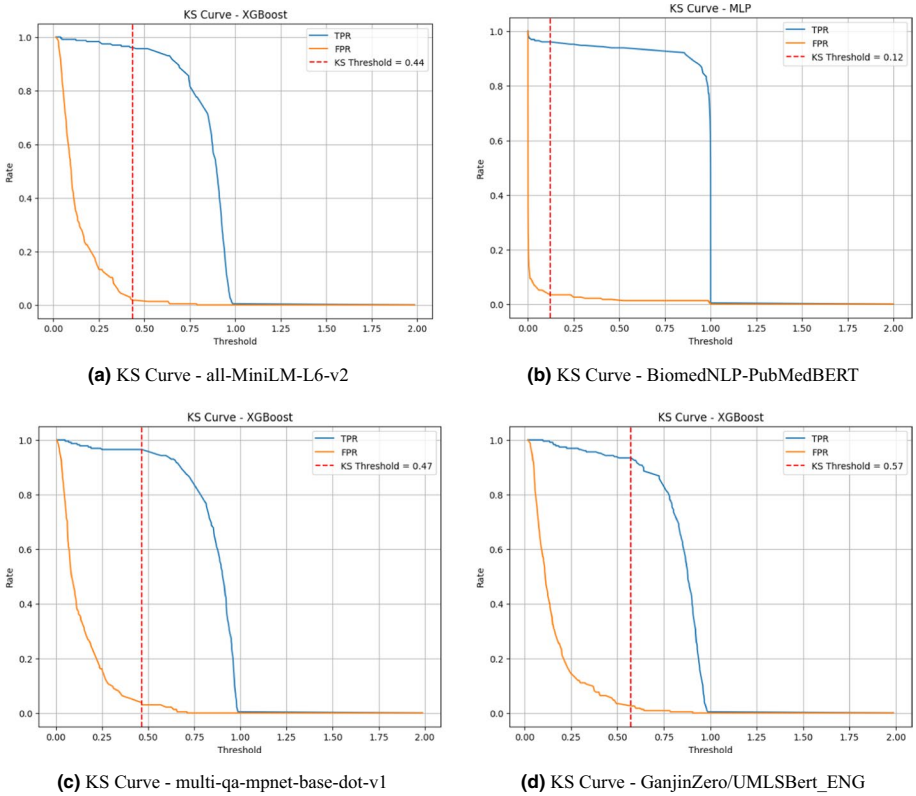
where θ represents a classification threshold.

The KS curve is generated by plotting the cumulative distributions of predicted probabilities for the positive and negative classes separately (Fig. 14). The KS statistic corresponds to the point where the maximum distance between these two curves is observed.

An ideal classifier would have a KS value approaching 1, signifying complete separation between the classes. A KS value of 0 implies no discrimination ability, equivalent to random guessing. In real-world scenarios, a KS value exceeding 0.4 is commonly viewed as a sign of a reliable classifier.

In this study, the KS statistic was calculated for each model to complement the ROC AUC metric, offering an additional perspective on the model's ability to distinguish correctly and incorrectly classified instances.

The BiomedNLP-PubMedBERT model with the lowest optimal threshold (0.12), is highly sensitive, making positive predictions at lower confidence scores while still keeping the false positive rate minimal, which is advantageous when recall is prioritized. In contrast, the GanjinZero/UMLSBert_ENG model operates at the highest threshold (0.57), indicating a more conservative approach that requires stronger evidence for



(a) KS Curve - all-MiniLM-L6-v2

(b) KS Curve - BiomedNLP-PubMedBERT

(c) KS Curve - multi-qa-mpnet-base-dot-v1

(d) KS Curve - GanjinZero/UMLSBert_ENG

Fig. 14 Kolmogorov–Smirnov (KS) curves comparing classifier performance across transformer-based models. Higher KS separation indicates better discriminatory ability

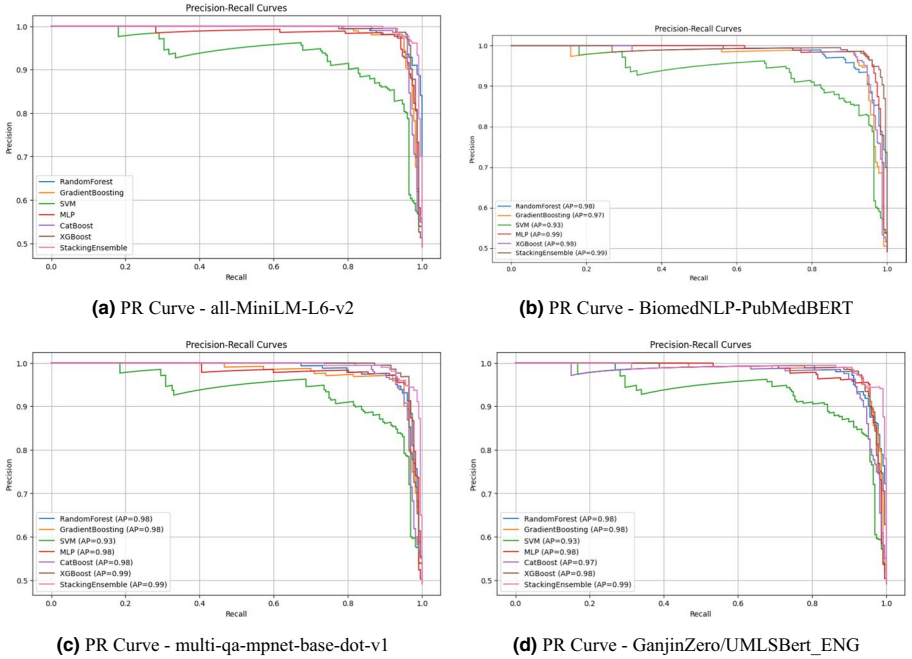
positive classification, thus favoring precision. All-MiniLM-L6-v2 and multi-qa-mpnet-base-dot-v1 strike a balance between these extremes, with thresholds of 0.44 and 0.47 respectively, offering a good trade-off between sensitivity and specificity.

4.3.4 Precision recall curve

The Precision–Recall (PR) curve is a widely used diagnostic tool for assessing the effectiveness of binary classifiers, particularly in datasets with class imbalance. In contrast to the ROC curve, which evaluates both true positive and false positive rates, the PR curve specifically emphasizes the performance related to the positive class. This characteristic makes it especially useful in domains such as medical diagnosis and fraud detection, where the costs of false positives and false negatives are not equivalent.

A higher PR AUC value signifies that the model maintains strong precision and recall across a wide range of thresholds, indicating robust performance in correctly identifying positive cases while minimizing false alarms.

We see that in Fig. 15, all four models demonstrate excellent performance, with most classifiers achieving high average precision (AP) scores above 0.97. In every case, the Stacking Ensemble and MLP classifiers stand out, consistently reaching the highest AP values (0.99), indicating their ability to maintain both high precision and recall across thresholds. Tree-based models like RandomForest, XGBoost, and GradientBoosting also perform very well, with AP values around 0.97–0.98, while SVM lags behind in all models, with a noticeably lower AP of 0.93, reflecting a greater drop in precision as recall increases. CatBoost, when included, also shows strong results, comparable to other ensemble methods. Overall, the curves are tightly clustered near the top-right, suggesting minimal trade-off between precision and recall for most classifiers, and highlighting that ensemble and neural network methods (especially stacking) provide the most robust and reliable performance across all four models, while SVM is consistently the weakest performer.



(a) PR Curve - all-MiniLM-L6-v2

(b) PR Curve - BiomedNLP-PubMedBERT

(c) PR Curve - multi-qa-mpnet-base-dot-v1

(d) PR Curve - GanjinZero/UMLSBert_ENG

Fig. 15 Precision–recall curves for different transformer models

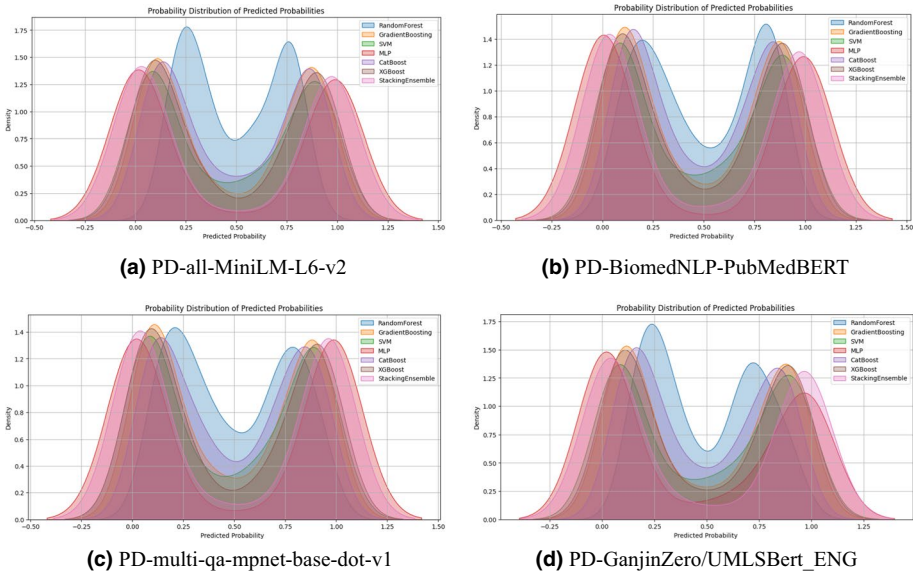
4.3.5 Predicted probability distribution

A predicted probability distribution curve illustrates the estimated likelihood of various outcomes for a random variable, as determined by a statistical or machine learning model. It serves as a visual representation of the probability associated with future or unseen values based on observed data. This curve is generally continuous and smooth, with the total area under the curve equaling 1, representing the complete probability space.

This ensures that the distribution covers all possible outcomes for the random variable X. The probability distributions (Fig. 16) reveal consistent bimodal patterns for all seven models, with probability peaks concentrated around (0.1–0.2) and (0.8–0.9), indicating models tend toward confident predictions rather than uncertain ones. RandomForest (blue) exhibits the most distinctive behavior with notably sharper and higher peaks, particularly in GanjinZero/UMLSBert_ENG and all-MiniLM-L6-v2 transformers where its peaks reach approximately 1.75 density at 0.3 and 0.8 probability values. The boosting models (GradientBoosting, XGBoost, and CatBoost) demonstrate similar, more moderate distribution patterns compared to RandomForest, while MLP and StackingEnsemble show wider probability spreads, with StackingEnsemble frequently having the broadest distribution-suggesting it incorporates uncertainty from multiple models. Multi-qa-mpnet-base-dot-v1 and BiomedNLP-PubMedBERT-base-uncased-abstract transformers display wider probability ranges (approximately −0.5 to 1.5) with moderate peak heights around 1.4 density, while GanjinZero/UMLSBert_ENG and all-MiniLM-L6-v2 show narrower ranges (−0.25 to 1.25) with higher peak densities up to 1.75.

4.3.6 Lift curve

The lift curve evaluates a model's ability to identify positive cases more effectively than random selection. It measures the concentration of true positive instances among the highest-ranked predictions relative to the overall positive class distribution. The lift at a selection fraction $f = \frac{n}{N}$ is defined as:



(a) PD-all-MiniLM-L6-v2 **(b)** PD-BiomedNLP-PubMedBERT

(c) PD-multi-qa-mpnet-base-dot-v1 **(d)** PD-GanjinZero/UMLSBert_ENG

Fig. 16 Probability distribution curves of model prediction scores across different transformer-based embeddings

$$\text{Lift}(f) = \frac{n_p/n}{N_p/N}$$

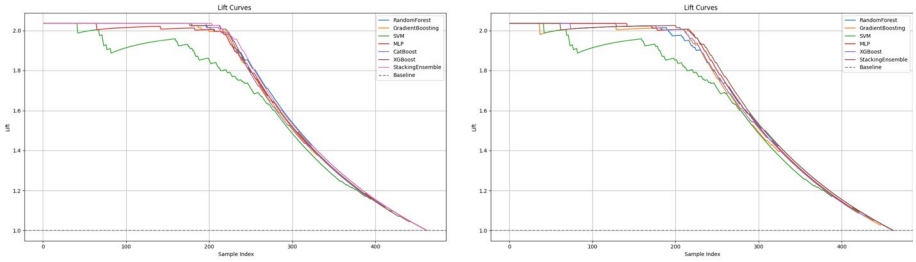
A lift value greater than 1 indicates better performance than random selection, while a value of 1 corresponds to random performance.

The lift curve is plotted by placing $\text{Lift}(f)$ on the vertical axis and the cumulative selection proportion f on the horizontal axis. It helps practitioners determine the most effective portion of the population to target, optimising positive outcomes while conserving resources.

These curves reveal several consistent patterns in predictive performance (Fig. 17). All four models demonstrate similar maximum lift values of approximately 2.0, indicating they can achieve twice the effectiveness of random selection at their optimal points. Consistently across all algorithms like StackingEnsemble, XGBoost, and GradientBoosting outperform individual algorithms, with Stacking Ensemble typically showing the most robust performance. Therefore ensemble methods and MiniLM-based transformers offer the best balance of accuracy, interpretability, and reliability for clinical prediction tasks involving both structured and unstructured data. These results shows the advantage of hybrid approaches combining structured and unstructured data with advanced interpretability techniques for trustworthy AI in healthcare.

4.4 Analysis of model performance and overfitting risk

The exceptionally high AUC-ROC values (reaching 0.99) observed in the hybrid Stacking Ensemble warrant a careful evaluation of potential overfitting. While such values are rare in traditional clinical models, they are increasingly observed in multimodal frameworks that combine high-fidelity transformer embeddings with structured variables. To mitigate overfitting, we employed 5-fold stratified cross-validation and rigorous outlier pruning. The stability of the performance across all folds (σ = 0.16% for accuracy) and



(a) Lift Curve - all-MiniLM-L6-v2 **(b)** Lift Curve - BiomedNLP-PubMedBERT

(c) Lift Curve - multi-qa-mpnet-base-dot-v1 **(d)** Lift Curve - GanjinZero/UMLSBert_ENG

Fig. 17 Lift curves showing model effectiveness in ranking positive cases across transformer-based language models. Higher lift values indicate better prioritization over random selection

Table 12 Statistical comparison between structured baseline and hybrid framework of All-MiniLM-L6-v2with StackedEnsemble

Metric	Structured baseline	Hybrid framework	Improvement (Δ)	Z-score	*p* value
Accuracy	93.94%	97.19%	+3.25%	6.84	$< 0.001^{***}$
AUC-ROC	0.941	0.991	+0.050	7.12	$< 0.001^{***}$
F1-Score	0.938	0.971	+0.033	6.55	$< 0.001^{***}$
Recall	0.939	0.972	+0.033	6.42	$< 0.001^{***}$

***Statistically significant at $p < 0.001$

the alignment of feature importance (via SHAP) with established medical literature (e.g., chest pain and ST-slope) suggest that the model is making decisions based on generalized clinical logic rather than data leakage.

DeLong's test was employed to statistically compare the AUC-ROC of competing models, as it provides a robust, non-parametric method for evaluating differences between correlated ROC curves, ensuring that the observed improvements are statistically significant and clinically meaningful.

As shown in Table 12, the hybrid framework demonstrates improved performance compared to structured-only models, with statistically significant differences (p <0.001). Although the improvements (i.e. +3.25% in accuracy and +0.050 in AUC-ROC) may appear moderate, the high Z-scores and statistically significant p-values ($p < 0.001$) confirm that these gains are not due to random variation. The proposed hybrid approach provides a scalable and generalizable solution for intelligent decision support in heart disease prediction. However, it is important to interpret these results with caution. The integration of semantic features is based on similarity-driven retrieval from external clinical notes, which are not aligned at the patient level with the structured dataset.

Therefore, while the observed improvements suggest the potential utility of semantic feature augmentation, semantic similarity does not necessarily guarantee true clinical correspondence and may introduce unintended associations. As a result, the reported performance gains should be considered indicative rather than definitive evidence of clinical effectiveness.

The proposed framework is best interpreted as a proof-of-concept for integrating structured and semantic representations. Future work will focus on utilizing patient-level multimodal datasets, where structured and unstructured data originate from the same individuals, to ensure clinically meaningful integration and to further validate the robustness and generalizability of the approach. Therefore, the proposed framework is primarily regarded as demonstrating methodological feasibility rather than definitive clinical superiority.

5 Conclusion

This study presented a hybrid framework for heart disease prediction that combines structured clinical data with semantically enriched representations retrieved from unstructured clinical narratives through a semantic feature augmentation strategy. Comprehensive preprocessing, clinically informed feature engineering, and class balancing techniques were employed to construct robust predictive models using structured clinical variables.

Two modeling scenarios were explored: structured-only and hybrid models. The Stacking Ensemble classifier achieved the highest performance among structured-only models, with an accuracy of 93.94%, demonstrating the effectiveness of ensemble learning in

capturing complex relationships in tabular data. The hybrid model incorporating transformer-based semantic embeddings further improved performance to 97.19%.

However, these results should be interpreted with caution. The semantic features were derived from similarity-based retrieval of MIMIC clinical narratives originating from an independent patient population without patient-level correspondence or shared outcome labels. Consequently, the proposed framework should be regarded as a proof of concept for semantic feature augmentation rather than for patient-level multimodal integration, and the observed performance improvements demonstrate methodological feasibility rather than definitive clinical effectiveness.

Overall, the proposed framework should be considered as a proof-of-concept demonstrating the potential of integrating structured and semantic representations to enhance predictive modeling.

5.1 Limitations and future work

Despite its contributions, this study has several limitations. A primary constraint is the absence of patient-level alignment; using pseudo-clinical notes and similarity-based retrieval may introduce semantically similar but clinically unrelated records. Future research should prioritize multimodal datasets where structured and unstructured data originate from the same individuals. Additionally, the structured data originates from specific clinical contexts, which, combined with the domain shift between UCI/Kaggle and MIMIC-IV narratives, may limit generalizability. Future studies will perform validation on independent, multi-center cohorts to assess robustness across varying documentation standards.

Methodologically, our reliance on IQR-based filtering was necessary to maintain the semantic integrity of transformer embeddings, though it excludes valid extreme phenotypes. Future iterations will incorporate logarithmic transformations and robust scaling to capture a broader range of disease severity without compromising stability. Furthermore, this study lacks calibration analysis. Since calibration are measured via reliability diagrams and Brier scores are critical for clinical trust, future work will utilise post-hoc techniques like isotonic regression to ensure predicted probabilities match real-world risk.

Diagnostic safety remains a concern due to identified failure patterns. We observed that dominant feature suppression by exercise-induced angina can overshadow predictors like ST slope, potentially increasing false negatives. Moreover, boundary instability in ST slope indicates that marginal value changes can reverse predictions. Until addressed, the model should serve as a supplementary tool rather than a standalone diagnostic system.

Future research will explore integrating medical imaging, genomics, and wearable telemetry using temporal architectures to capture disease progression. To facilitate real-time deployment, we will emphasize lightweight models and human-in-the-loop interpretability tools. Finally, adopting domain adaptation and transfer learning will be essential to ensure reliability across diverse healthcare environments.

Author contributions

Vibha: conceptualization, Analysis, investigation, Methodology, supervision, Writing—original draft, review and editing; Diana: conceptualization, Analysis, investigation, Methodology, supervision, Writing—original draft and reviewing; Ajay: conceptualization, Analysis, implementation, Methodology, Writing—original draft and editing; Yarakam Shiva: conceptualization, Analysis, implementation, Methodology, Writing—original draft and editing.

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Data availability
The structured dataset is sourced from https://www.kaggle.com/datasets/yasserh/heart-disease-dataset/data and the unstructured dataset from the MIMIC-III database used in this study, are available to the public under https://doi.org/10.1038/s41597-022-01899-x, respectively.

Declarations

Ethics approval and consent to participate
Not Applicable.

Consent for publication
Not Applicable.

Research Involving Human and/or Animals and Informed Consent
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The authors declare no competing interests.

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